
Calibrated Is Not Actionable: Physics-Guided Posterior Distillation for Methane Source Attribution

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Abstract

The physics allows 7 meters; regulators ask for 50; a calibrated neural localizer delivers 94. On a methane-configured testbed (500 m site, Gaussian plume under realistic meandering wind, 4–12 fixed masts) where the *exact* Bayesian posterior of every scenario is computable, a conformal-wrapped deep localizer attains verified 90% coverage while its median search region has a 94 m equivalent radius — honestly calibrated, nearly twice the U.S. EPA super-emitter actionability radius, and an order of magnitude above the 7 m information limit. We diagnose this gap by running the *same conformal audit on every stage of the pipeline*: the sensors support 7 m; the tempered exact-posterior teacher used for distillation supports 11.7 m; the trained student delivers 74 m — and neither sharper targets (87 m) nor a spatially aware transport loss (243 m) help. The bottleneck is a computation, not supervision: set encoders cannot perform per-cell matched filtering internally. We therefore hand the network the classical matched-filter statistics of the known forward model as input maps, through a zero-initialized residual head, while distilling the tempered exact posterior — a drop-in recipe for any localizer. Across three architecture families (DeepSets, GNN, Set Transformer) and three seeds each, the recipe brings every model under the 50 m bar (44–49 m; coverage 0.89–0.91; sharper on 91–96% of scenarios, paired $p < 10^{-200}$), while collapsing the spread between architectures. Calibration alone does not make ML monitoring actionable; auditing against the physics limit, and giving the network the physics it cannot rediscover, can.

1 Introduction

Methane is the highest-leverage near-term climate lever: roughly $80\times$ the 20-year warming potency of CO_2 , a $\sim$ decade atmospheric lifetime, and a heavy concentration of emissions in a small number of fixable oil-and-gas super-emitters. Detection is increasingly solved — UNEP’s Methane Alert and Response System issued over a thousand satellite notifications in its first two years — but only $\sim 1\%$ led to documented action [UNEP, 2024]. The missing step is *attribution*: converting a detection into a source location precise enough, and trustworthy enough, to dispatch a crew or attach a reporting obligation. The U.S. EPA Super-Emitter Program operationalizes “precise enough” as a 50 m radius [U.S. EPA, 2024]; the EU methane regulation and OGMP 2.0 impose analogous source-level accounting duties [European Union, 2024]. We treat 50 m as a concrete engineering target.

Continuous monitoring with sparse networks of fixed point sensors is the tier of the monitoring stack where attribution must happen: a handful of cheap masts around a facility, always on, catching the plume as it sweeps across them under naturally meandering wind. A learned localizer can map readings to a heatmap of candidate sources, and split conformal prediction [Angelopoulos & Bates,

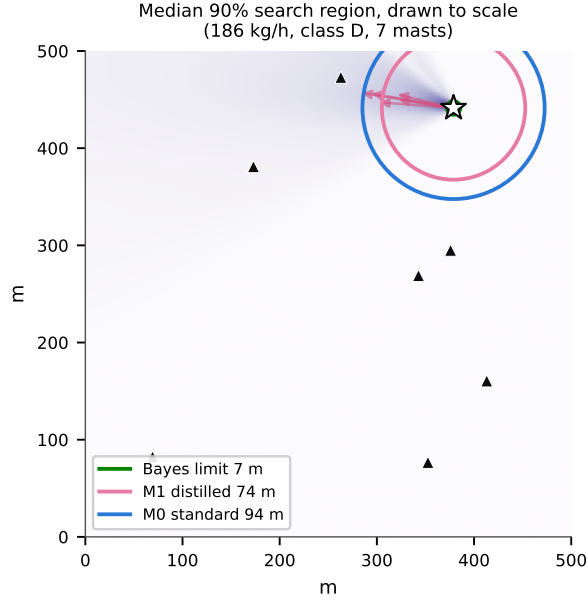


Figure 1: **The problem, to scale.** A wind-swept methane plume (purple; meander arrows), 7 masts (black), and the true source (star) on a 500 m site. Circles: the median 90% search region the data supports (Bayes limit, 7 m), what a standard calibrated localizer delivers (94 m), and the EPA 50 m actionability radius sits between them. Calibration is not actionability.

2023] can convert the heatmap into a region guaranteed to contain the true source 90% of the time. Coverage, however, is a statement about frequency, not usefulness: a region can be honestly 90% and still be far too large to act on. This paper is about measuring that gap exactly, diagnosing its mechanism, and closing it.

Contributions. (1) An *exact-posterior audit*: a testbed whose closed-form physics makes the true Bayesian posterior of every scenario computable, so the efficiency of any calibrated localizer can be measured against the information limit, in meters, against the regulatory radius. (2) A *diagnosis by elimination*, including a new one-run diagnostic — conformalizing the *teacher* of a distillation setup — that decomposes the calibrated-sharpness gap into a supervision ceiling versus a student approximation failure. (3) *Physics-guided posterior distillation*: a drop-in recipe (matched-filter input maps + zero-initialized residual head + tempered exact-posterior distillation) that brings three different localizer architectures under the 50 m actionability radius with the coverage guarantee intact. We claim no new conformal algorithm, no new architecture, no coverage under distribution shift, and no field deployment.

2 Testbed: exact posteriors as an instrument

CH4-T places a source of unknown location and rate (10–500 kg/h, log-uniform) on a 500×500 m site sensed by 4–12 masts (ppm readings, 0.2–3 ppm noise, dropout). The plume is a steady 3-D Gaussian under Pasquill–Gifford stability (known class, Briggs open-country dispersion), driven quasi-statically by a *known, fluctuating* wind record (mean 1–8 m/s with AR(1) meander in direction and speed) — the wind that on-site anemometry measures. As the wind veers, the plume sweeps the masts; time-of-arrival information localizes sources that a constant wind cannot (weakly identifiable scenarios drop from 57% to 6%).

Because the response is closed-form and linear in the rate, the *exact* posterior over a 64×64 grid (cell-level attribution; sources on the grid) is computable per scenario: a Gaussian likelihood whose rate marginalization is done by an adaptive quadrature, validated by a blocking self-calibration gate — conformal prediction applied to the exact posterior itself must recover nominal coverage (it does: 0.95), a gate that caught a real forward-model bug during development. The exact posterior

serves twice: as the *reference* that turns “how sharp could a calibrated region be” into a number, and as a *teacher*. All models below use identical data (10k/1k/2k/2k train/val/calib/test), identical D4 symmetry augmentation (without which set encoders memorize), and an identical randomized-HPD split-conformal wrapper at verified 90% coverage, computed in numerically safe tail-mass form, in float64. Region sizes are reported as the radius of the circle of equal area; the domain is 25 ha.

3 Diagnosis: interrogate every stage with the same instrument

Calibrated $\neq$ actionable. A DeepSets localizer trained on smoothed one-hot labels (M0) is well calibrated under conformal (0.89) yet its median region radius is 94 m — $13\times$ the Bayes limit in radius, $\sim 180\times$ in area. The information was in the readings; the model could not extract it.

The teacher is not the bottleneck. Distilling the exact posterior — same network, same data, only the target changes — requires tempering: the raw near-delta posterior fails outright (134 m, *worse than no physics*), while a 0.75-cell Gaussian blur of it works (74 m). It is tempting to blame the blur for the remaining gap. Our diagnostic says otherwise: run the conformal audit *on the tempered teacher itself*. The teacher supports 11.7 m. A perfect student of the existing lesson would already be $6\times$ sharper than the actual student; the binding constraint is the student’s failure to match its teacher, not the lesson.

Supervision cannot fix it. Two controlled attempts to repair training — annealing the temper toward the raw posterior (0.75 $\rightarrow$ 0.15 cells: 87 m, worse), and replacing cross-entropy with a debiased Sinkhorn (optimal-transport) divergence to the raw posterior with coarse-to-fine ε -scaling (243 m) — both fail while remaining calibrated. (We scope the transport result to our implementation; it does not preclude better OT variants.)

Verdict: a computation deficit. The evidence indicates the encoder cannot represent the required computation: identifying the source demands correlating each of 4,096 candidate cells’ predicted space–time signature against all readings — per-cell matched filtering — which a pooled set representation cannot carry. No target and no loss teaches a network what its forward pass cannot compute.

4 Method: physics-guided posterior distillation

The forward model is known; the computation the network lacks has a formula. The recipe hands it over and lets learning do the rest. For any base localizer:

(1) Physics maps. From the observed readings y , wind record, and stability class, compute per cell c the sufficient statistics of the linear-Gaussian model, $a_c = \sum g_c y$ and $b_c = \sum g_c^2$ (with g_c the unit-rate response), and feed two 64×64 channels: the matched-filter statistic $z_c = a_c / (\sigma \sqrt{b_c})$ — classical detection theory — and a sensitivity map $\log b_c$. Milliseconds, closed-form, computed from the same inputs the network already receives (no extra information; a computation).

(2) Zero-initialized residual head. A 3-layer conv head reads the maps and adds to the base model’s logits. Zero-initialized final layer: at step 0 the model *is* the baseline; training opens the physics channel only where it reduces loss. The recipe is drop-in for any architecture and the comparison is controlled by construction.

(3) Tempered posterior distillation. Cross-entropy to the 0.75-cell-blurred exact posterior. The maps supply evidence; the teacher teaches how to weigh evidence into a calibrated probability shape (rate prior, noise level, multi-modality). **(4)** The unchanged conformal wrapper certifies the result.

Why not just use the teacher?

On this testbed one should — the exact posterior is our measuring instrument, computable by construction. In deployment it vanishes: realistic dispersion (terrain, transients) has no closed form, and simulator inversion costs hours per event [Newman et al., 2024] where monitoring needs seconds across thousands of sites. Amortization is the accepted answer; what was missing is what it *costs*. And “the teacher is just part of the dataset” is our thesis, not an objection: every simulated training set already contains its own uncertainty labels; standard training throws them away. The recipe is how to get them back — and what must be handed over (a matched filter) before they become learnable.

5 Results: one recipe, three architectures, all under the bar

Table 1: Median 90% conformal region radius (m) on 2,000 test scenarios; range over 3 seeds. Every cell is calibrated (coverage 0.887–0.910). References: Bayes limit 7.0 m, tempered-teacher ceiling 11.7 m, **EPA actionability** 50 m. Baseline columns for GNN/Set Transformer reproduce companion architecture work on the identical harness.

Architecture	Labels only	+ Distillation	+ Full recipe
DeepSets (4.9M)	94–98	73–74	48.1–49.3
GNN (5.6M)	55–58	49–52	43.6–44.3
Set Transformer (6.5M)	69–75	56–60	44.7–48.9

Three observations. **(i) Universality:** nine of nine seed-runs of the recipe land under the 50 m radius; each improves its architecture’s distilled baseline on 91–96% of scenarios (paired Wilcoxon on log region size, $p < 10^{-200}$). **(ii) Not a crutch for weak encoders:** the best architecture (GNN, whose relational structure already computes pairwise mast–wind geometry) also improves, from 49–52 to 43.6–44.3 m. **(iii) Architecture stops mattering:** the labels-only spread across architectures (94 vs. 56 m) collapses under the recipe (49 vs. 44 m) — supervision and inputs, not encoder design, were the binding constraint. Point accuracy is essentially unchanged throughout; what the recipe changes is the *shape of uncertainty*, which is what the region size measures. Inference cost is unchanged ($\sim$ ms; the maps add milliseconds of closed-form computation).

6 Limitations and outlook

CH4-T is an instrument, not a deployment: steady release, flat terrain, known stability class, quasi-static plume sensed in 2-D, *exactly known* wind, marginal in-distribution coverage. The residual factor of ~ 6 between the recipe (44–49 m) and the Bayes limit (7 m) is unexplained headroom. The sufficiency of the input maps is exact only under the testbed’s linear-Gaussian forward model; under misspecification they are informative features, not sufficient statistics. We have not compared against continuous density estimators (normalizing-flow posterior estimation), and our transport -loss null is implementation-scoped. Measured-wind uncertainty — anemometer error, the deployment reality — is the immediate next act: the Bayes-correct response (wind-marginalized teachers and ensemble physics maps) is implemented and under evaluation. The path to field claims runs through controlled-release data (METEC-style); what this study establishes is the reliability machinery — audit against the physics limit, diagnose by elimination, hand over what cannot be learned — and that with it, calibrated ML attribution can reach the radius regulators call actionable.

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